1 · DIMENSIONALIDAD

Covarianza estimada (d. 5)	$\hat{\mathbf{C}} = \frac{1}{N}\sum_i (\mathbf{x}_i - \hat{\boldsymbol{\mu}})(\mathbf{x}_i - \hat{\boldsymbol{\mu}})^T$, $\quad \text{rank}(\hat{\mathbf{C}}) \le \min(d, N-1) \Rightarrow$ singular si $N \le d$
Parámetros de $\mathbf{C}$	$\frac{d(d+1)}{2}$ por clase (crece como d^2)
Dos gaussianas, misma $\mathbf{C}$, equiprobables (d. 8–9)	$\Pr(e) = \frac{1}{\sqrt{2\pi}}\int_{r/2}^{\infty} e^{-u^2/2}du = Q(r/2), \quad r^2 = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T\mathbf{C}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$
Características independientes	$r^2 = \sum_{i=1}^{d}\left(\frac{\mu_{i,1}-\mu_{i,2}}{\sigma_i}\right)^2 = \sum_i D_i'^2$; relevantes y $d \to \infty \Rightarrow \Pr(e) \to 0$ (parámetros conocidos)
Valores de Q	$Q(0) = 0{,}5 \quad Q(0{,}5) = 0{,}309 \quad Q(1) = 0{,}159 \quad Q(2) = 0{,}023 \quad Q(3) = 0{,}0013$

2 · TRANSFORMACIÓN LINEAL (d. 13)

Reducción	$\mathbf{z}_k = \mathbf{W}^T(\mathbf{x}_k - \boldsymbol{\alpha}), \quad \mathbf{x} \in \mathbb{R}^d,\ \mathbf{z} \in \mathbb{R}^{d'},\ \mathbf{W} \in \mathbb{R}^{d\times d'},\ d' < d$
Reconstrucción (d. 14)	$\hat{\mathbf{x}}_k = \mathbf{W}\mathbf{z}_k + \boldsymbol{\alpha}, \qquad \mathbf{e}_k = \mathbf{x}_k - \hat{\mathbf{x}}_k$
Componente j	$z_{k,j} = \mathbf{w}_j^T(\mathbf{x}_k - \boldsymbol{\alpha})$ = coordenada sobre $\mathbf{w}_j$ si $\|\mathbf{w}_j\| = 1$

3 · PCA (d. 14–25)

Criterio	$J = \sum_{k=1}^{N}\|\mathbf{W}\mathbf{z}_k + \boldsymbol{\alpha} - \mathbf{x}_k\|_2^2$ (mínimo)
Solución (d. 15)	$\boldsymbol{\alpha} = \mathbf{m} = \frac{1}{N}\sum_k \mathbf{x}_k, \quad \mathbf{S} = \sum_k(\mathbf{x}_k - \mathbf{m})(\mathbf{x}_k - \mathbf{m})^T, \quad \mathbf{S}\mathbf{w}_j = \lambda_j\mathbf{w}_j, \quad \mathbf{W} = [\mathbf{w}_1 \cdots \mathbf{w}_{d'}]$ con los d' mayores λ_j
$\mathbf{S}$ y covarianza	$\mathbf{S} = (N-1)\hat{\mathbf{C}}_{\text{insesgada}} = N\hat{\mathbf{C}}_{\text{MV}}$: mismos autovectores
Propiedades de $\mathbf{S}$ (p. 18)	$\mathbf{S} = \mathbf{S}^T$, $\mathbf{v}^T\mathbf{S}\mathbf{v} \ge 0$, $\lambda_j \ge 0$, autovectores ortogonales, $\lambda_j = \sum_k z_{k,j}^2$, $\mathbf{W}^T\mathbf{S}\mathbf{W} = \text{diag}(\lambda_j)$
$\mathbf{z}_k$ óptimo, $\mathbf{W}$ cualquiera (d. 17)	$\mathbf{z}_k = (\mathbf{W}^T\mathbf{W})^{-1}\mathbf{W}^T(\mathbf{x}_k - \mathbf{m}), \quad \hat{J} = -\sum_k(\mathbf{x}_k - \mathbf{m})^T\mathbf{P}(\mathbf{x}_k - \mathbf{m})$
Proyector (d. 18)	$\mathbf{P} = \mathbf{W}(\mathbf{W}^T\mathbf{W})^{-1}\mathbf{W}^T = \mathbf{U}_L\mathbf{U}_L^T$ (simétrico, idempotente) ⇒ se toma $\mathbf{W}^T\mathbf{W} = \mathbf{I}$
Traza (d. 19)	$\sum_k(\mathbf{x}_k - \mathbf{m})^T\mathbf{W}\mathbf{W}^T(\mathbf{x}_k - \mathbf{m}) = \text{tr}(\mathbf{W}^T\mathbf{S}\mathbf{W}) = \sum_{i=1}^{d'}\mathbf{w}_i^T\mathbf{S}\mathbf{w}_i$
Lagrangiano	$\sum_i \mathbf{w}_i^T\mathbf{S}\mathbf{w}_i + \sum_i \lambda_i(E - \mathbf{w}_i^T\mathbf{w}_i) \Rightarrow \mathbf{S}\mathbf{w}_i = \lambda_i\mathbf{w}_i$ (más ortogonalidad entre columnas)
Error mínimo (d. 20)	$J_{\min} = \sum_k\|(\mathbf{I} - \mathbf{W}\mathbf{W}^T)(\mathbf{x}_k - \mathbf{m})\|^2 = \text{tr}(\mathbf{S}) - \text{tr}(\mathbf{W}^T\mathbf{S}\mathbf{W}) = \sum_{i=d'+1}^{d}\lambda_i$
Hiperelipsoide (d. 23)	$(\mathbf{x} - \mathbf{m})^T\mathbf{S}^{-1}(\mathbf{x} - \mathbf{m}) = K \iff \sum_i \frac{z_i^2}{K\lambda_i} = 1$: semiejes $\sqrt{K\lambda_i}$ en las direcciones $\mathbf{w}_i$
Restar la media (d. 24)	$\sum_k \mathbf{x}_k\mathbf{x}_k^T = \mathbf{S} + N\mathbf{m}\mathbf{m}^T$
Eigenfaces (d. 22)	$\tilde{\mathbf{x}} = \mathbf{W}\mathbf{z} + \mathbf{m} = \mathbf{m} + \sum_{j=1}^{d'} z_j\mathbf{w}_j$

4 · MDA (d. 26–37)

Medias (d. 28)	$\mathbf{m}_i = \frac{1}{N_i}\sum_{\mathbf{x}\in D_i}\mathbf{x}, \qquad \mathbf{m} = \frac{1}{N}\sum_{\mathbf{x}}\mathbf{x} = \frac{1}{N}\sum_{i=1}^{c} N_i\mathbf{m}_i$
Dispersiones (d. 30)	$\mathbf{S}_{C,i} = \sum_{\mathbf{x}\in D_i}(\mathbf{x}-\mathbf{m}_i)(\mathbf{x}-\mathbf{m}_i)^T, \quad \mathbf{S}_C = \sum_i \mathbf{S}_{C,i}, \quad \mathbf{S}_B = \sum_i N_i(\mathbf{m}_i - \mathbf{m})(\mathbf{m}_i-\mathbf{m})^T$
Descomposición	$\mathbf{S}_T = \sum_{\mathbf{x}}(\mathbf{x}-\mathbf{m})(\mathbf{x}-\mathbf{m})^T = \mathbf{S}_C + \mathbf{S}_B$ (cruzados nulos: $\sum_{\mathbf{x}\in D_i}(\mathbf{x}-\mathbf{m}_i)=\mathbf{0}$)
En el espacio de $\mathbf{z}$ (d. 31)	$\mathbf{z} = \mathbf{W}^T\mathbf{x}: \quad \mathbf{S}_C \to \mathbf{W}^T\mathbf{S}_C\mathbf{W}, \quad \mathbf{S}_B \to \mathbf{W}^T\mathbf{S}_B\mathbf{W}$
Volumen del elipsoide	$V = V_d\lvert\mathbf{S}\rvert^{1/2}K^{d/2}, \quad V_d = \frac{\pi^{d/2}}{(d/2)!}$ (par), $\frac{2^d((d-1)/2)!}{d!}\pi^{(d-1)/2}$ (impar)
Criterio (d. 33)	$\mathbf{W} = \arg\max_{\mathbf{W}} \frac{\lvert\mathbf{W}^T\mathbf{S}_B\mathbf{W}\rvert}{\lvert\mathbf{W}^T\mathbf{S}_C\mathbf{W}\rvert} \quad (= D^2,\ D = \text{cociente de volúmenes})$
Solución	$\mathbf{S}_B\mathbf{w}_j = \sigma_j\mathbf{S}_C\mathbf{w}_j \iff \mathbf{S}_C^{-1}\mathbf{S}_B\mathbf{W} = \mathbf{W}\mathbf{\Sigma}$; los d' mayores σ_j
Valor (d. 34)	$\frac{\lvert\mathbf{W}^T\mathbf{S}_B\mathbf{W}\rvert}{\lvert\mathbf{W}^T\mathbf{S}_C\mathbf{W}\rvert} = \lvert\mathbf{\Sigma}\rvert = \prod_{i=1}^{d'}\sigma_i$
Dimensión máxima	$d' \le \min(d, c-1)$: $\operatorname{rank}(\mathbf{S}_B) \le c-1$, $\operatorname{rank}(\mathbf{S}_C) = d$ (exige $N-c \ge d$)
Dos clases (ej. 4)	$\mathbf{S}_B = \frac{N_1N_2}{N}(\mathbf{m}_1-\mathbf{m}_2)(\mathbf{m}_1-\mathbf{m}_2)^T, \quad \mathbf{w} \propto \mathbf{S}_C^{-1}(\mathbf{m}_1-\mathbf{m}_2)$
$\mathbf{S}_T$ en lugar de $\mathbf{S}_C$	mismos $\mathbf{w}_j$; autovalores $\frac{\sigma_j}{1+\sigma_j}$
Por grupos (d. 35)	K grupos de características ⇒ $d' = (c-1)K$
Clasificador en $\mathbf{z}$ (d. 37)	$\boldsymbol{\mu}_i = \mathbf{W}^T\mathbf{m}_i, \qquad \mathbf{C}_i = \frac{1}{N_i}\mathbf{W}^T\mathbf{S}_{C,i}\mathbf{W}$

5 · EJERCICIOS CLAVE

Ej. 1	$\mathbf{w}_i^T\mathbf{S}_C\mathbf{w}_j = \delta_{ij} \Rightarrow \tilde{\mathbf{S}}_C = \mathbf{I},\ \tilde{\mathbf{S}}_B = \operatorname{diag}(\lambda_j),\ J = \prod\lambda_i$; invariante a $\mathbf{y} = \mathbf{Q}\mathbf{D}\mathbf{W}^T\mathbf{x}$
Ej. 2 y 5	$\max_{\lVert\mathbf{w}\rVert=1}\mathbf{w}^T\mathbf{S}\mathbf{w} = \lambda_{\max}$; con $\mathbf{W}^T\mathbf{W} = \mathbf{I}$: $\max\operatorname{tr}(\mathbf{W}^T\mathbf{S}\mathbf{W}) = \sum_{i\le d'}\lambda_i$
Ej. 3	$\sum_i(\mathbf{m}_i-\mathbf{m}) = \mathbf{0} \Rightarrow \operatorname{rank}(\mathbf{S}_B) \le c-1$; $\operatorname{rank}(\mathbf{M}\mathbf{M}^T) = \operatorname{rank}(\mathbf{M})$
Ej. 4	$\nabla\frac{\mathbf{w}^T\mathbf{S}_B\mathbf{w}}{\mathbf{w}^T\mathbf{S}_C\mathbf{w}} = \mathbf{0} \Rightarrow \mathbf{S}_B\mathbf{w} = \lambda\mathbf{S}_C\mathbf{w}$, λ = el propio cociente

6 · PCA FRENTE A MDA (d. 43)

PCA	No supervisado · reconstruye · $d' \le d$ · no garantiza separar clases
MDA	Supervisado · separa clases · $d' \le \min(d, c-1)$ · necesita $\mathbf{S}_C$ invertible
No lineal (d. 38–42)	t-SNE (visualizar, conserva vecindades) · autoencoders (lineal + MSE ≡ subespacio PCA)